Weight Is a Force of Gravity on a Mass

SYNOPSIS

In everyday speech “weight” is a number on a bathroom scale, reported in pounds — a unit of *mass*. In physics it is something else entirely: weight is a **force**, measured in newtons, the pull of a planet on a mass. Mass belongs to the object and travels with it. Weight does not: it is what a *particular planet* does to that object. This note measures the relationship between the two, finds that it is a straight line, and shows why the slope of that line is the reason a feather and a bowling ball fall together in a vacuum.

BEFORE YOU READ ON — PREDICT

You hold a **20 kg** mass in one hand and a **10 kg** mass in the other, and release them at the same instant. Gravity pulls *twice as hard* on the 20 kg mass — that much is not in dispute.
Does it reach the floor first?

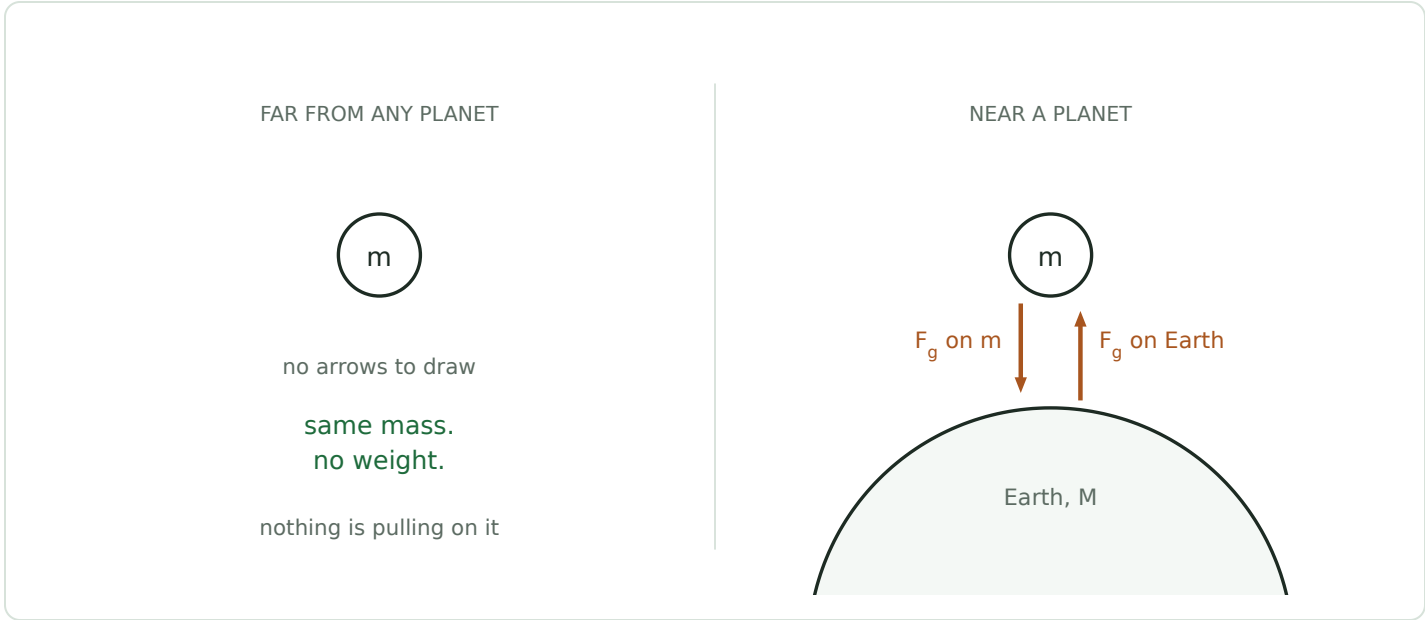
Commit to yes or no, and write one sentence saying why, before you read further.

PERFORMANCE EXPECTATIONS

1. **Distinguish** mass from weight by unit, by definition, and by what happens to each when the object is moved to another planet.
2. **Measure** weight against mass with a spring scale, plot the two, and **interpret** the slope of the line, including its unit.
3. **Explain** why every mass near a planet falls with the same acceleration, in terms of force *and* inertia.
4. **Identify** the third-law partner of an object’s weight, and state which body it acts on.
5. **Calculate** weight, mass, normal force, or g given the other quantities, on Earth or on any other planet.

MOVEMENT 1 — WEIGHT IS NOT A PROPERTY OF THE OBJECT

Where does weight come from?



1. A book rests on a table. Force given: **the table pushes up on the book**.

   partner _____ acts on _____

2. The same book on the same table. Force given: **the Earth pulls down on the book**.

   partner _____ acts on _____

3. The book is now in free fall, halfway to the floor. Force given: **the Earth pulls down on the book**.

   partner _____ acts on _____

   Has your answer changed from the previous question? Should it have?

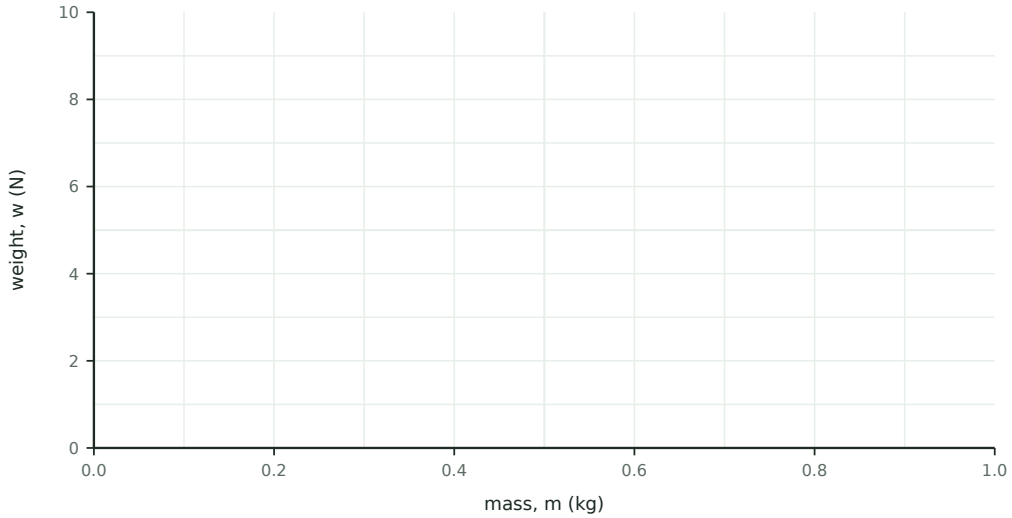
MOVEMENT 2 — MEASURE IT

Weight against mass

Hang a known mass on a spring scale and read the force it registers. Repeat with different masses. You are asking the scale a simple question: when the mass goes up, what does the pull do?

1. Zero the spring scale with nothing hanging from it.
2. Hang each mass in turn and let it come to rest before you read the needle. Record the reading in **newtons**.
3. Fill in the table. Keep the readings to the precision the scale actually gives you.

MASS, M (KG)	WEIGHT, W (N)
0.1	
0.2	
0.5	
0.7	
0.9	



Plot weight against mass. Draw the best straight line through your points — and think about whether that line is entitled to pass through the origin.

1. Is the plot linear? What does that tell you about how weight depends on mass?

2. Find the slope, and give its unit. Do not skip the unit — it is the whole point of the next question.

3. What does the slope represent? Where have you seen that number before?

THE RESULT

$w = F_g = mg$

MOVEMENT 3 — WHY EVERYTHING FALLS TOGETHER

Your prediction, settled

Eq. 1 $F_g = ma$ — the only force on a freely falling mass is its weight

Eq. 2 $a = \frac{F_g}{m} = \frac{mg}{m} = g$ — the mass cancels — the acceleration does not depend on it

4. “You drop a 20 kg and a 10 kg mass. The 20 kg mass experiences twice as large a force of gravity as the 10 kg mass, but they still accelerate at the same rate, 9.8 m/s².” True or false?

5. Give the reason for your choice in one sentence. Your sentence must mention **both** force and inertia — an answer that mentions only one of them is not an explanation.

6. Go back to your prediction in the box at the top. Was it right? If not, name the step in the reasoning where it went wrong.

7. Compare this with the guess you wrote in N2. Which part did you already have, and which part were you missing?

8. Fire the ball **twice as fast**. Do you still hit the target? What changes about *where* and *when* you hit it?

9. Now break it. Name a change to the demonstration that would make the shot miss — and say which of the two moves above your change destroys.

MOVEMENT 4 — WEIGHT, NORMAL FORCE, AND OTHER PLANETS

An object at rest on a level floor

10. An obese cat with a mass of 46 kg rests on the floor. Find its weight and the magnitude of the normal force on it, and draw its free-body diagram.

11. A cat weighs 57.3 N. What is its mass? Resting on a floor, what normal force acts on it? If the cat is sent to another planet, what is its mass there?

12. On the Moon, a person of mass 70 kg weighs 112 N. What is the acceleration due to gravity on the Moon?

13. A 65 kg astronaut weighs 638 N on Earth. What is the astronaut's **mass** on Mars, where $g = 3.4\ m/s^2$? What is the astronaut's **weight** there?

THE HABIT TO BUILD

Mass is given in kilograms and never changes. Weight is given in newtons and changes with the planet. If a quantity in a problem is in kg it is a mass; if it is in N it is a force. Check the unit before you decide which one you have been handed.

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